

Problem 97. The positive integers $a_0, a_1, a_2, \dots, a_{3030}$ satisfy

$$2a_{n+2} = a_{n+1} + 4a_n \quad \text{for } n = 0, 1, 2, \dots, 3028.$$

Prove that at least one of the numbers $a_0, a_1, a_2, \dots, a_{3030}$ is divisible by 2^{2020} .

Problem 98. A permutation of the integers $1, 2, \dots, m$ is called *fresh* if there exists no positive integer $k < m$ such that the first k numbers in the permutation are $1, 2, \dots, k$ in some order. For example, if $m = 4$, then the permutation $(3, 1, 4, 2)$ is fresh, whereas the permutation $(2, 3, 1, 4)$ is not.

Let f_m be the number of fresh permutations of the integers $1, 2, \dots, m$.

Prove that $f_n \geq n \cdot f_{n-1}$ for all $n \geq 3$.

Problem 99. Consider the triangle ABC with $\angle BCA > 90^\circ$. The circumcircle Γ of ABC has radius R . There is a point P in the interior of the line segment AB such that $PB = PC$ and the length of PA is R . The perpendicular bisector of PB intersects Γ at the points D and E .

Prove that P is the incentre of triangle CDE .

Problem 100. Let $m > 1$ be an integer. A sequence $a_1, a_2, a_3, \dots$ is defined by $a_1 = a_2 = 1$, $a_3 = 4$, and for all $n \geq 4$,

$$a_n = m(a_{n-1} + a_{n-2}) - a_{n-3}.$$

Determine all integers m such that every term of the sequence is a square.